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CO5-AT2-Application Based Problem Solving

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QUESTIONS

Q1

Employee Management and Payroll Application using Java AWT and MySQL

25 marks

0/1 answered

Q1

Employee Management and Payroll Application using Java AWT and MySQL

25 marks

A medium-sized organization currently maintains employee information using spreadsheets and manually prepared records. As the number of employees increases, the HR department faces difficulties in maintaining accurate employee details, updating salary information, tracking department-wise employees, and generating payroll-related information. Duplicate records, inconsistent data, and difficulty in retrieving employee information have also become common problems. The organization therefore proposes to develop a desktop-based Employee Management and Payroll Application using Java AWT for the graphical user interface and MySQL for persistent data storage.

The proposed application should provide a user-friendly interface through which an authorized HR staff member can add, view, search, update, and delete employee records. Each employee should have details such as Employee ID, Name, Gender, Date of Birth, Contact Number, Email, Address, Department, Designation, Date of Joining, Basic Salary, Allowances, Deductions, and Employment Status. The system should maintain appropriate relationships between employee and

department information and ensure that Employee ID and other required fields are validated before storing the data.

The application should also provide a salary/payroll module that calculates an employee's gross salary and net salary based on basic salary, allowances, deductions, and applicable rules. The system should allow HR personnel to search employees based on Employee ID, department, designation, or employment status. The application should prevent invalid operations such as duplicate Employee IDs, missing mandatory fields, invalid salary values, and attempts to update or delete non-existing employees. All database operations must be performed through JDBC, and appropriate exception handling must be incorporated to manage database and application-level errors.

The application should be designed in a modular manner so that additional functionality such as attendance, leave management, performance tracking, or employee appraisal can be incorporated in the future without completely redesigning the application.

Tasks to be Performed

1. Requirement Analysis and Database Design

Analyze the given problem and identify the functional and non-functional requirements. Design the MySQL database by identifying appropriate tables, attributes, primary keys, foreign keys, constraints, and relationships required for the Employee Management Application.

2. GUI Design using Java AWT

Design and implement suitable AWT-based screens for:

Employee Registration

Employee Search

Employee Update

Employee Delete

Employee List/View

Payroll/Salary Calculation

Use appropriate AWT components such as Frame, Panel, Label, TextField, Choice, Checkbox, Button, TextArea, Dialog, etc., and implement suitable event-handling mechanisms.

3. JDBC and CRUD Implementation

Establish connectivity between the Java application and MySQL using JDBC. Implement complete CRUD



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4. Payroll and Validation Module

Implement a payroll module to calculate:

Gross Salary = Basic Salary + Allowances

Net Salary = Gross Salary – Deductions

Display the employee's salary details through the AWT interface. Implement validation for mandatory fields, numeric salary values, duplicate Employee IDs, invalid input, and other appropriate business rules.

5. Testing, Exception Handling and Evaluation

Test the complete application using appropriate test cases covering:

New employee registration

Duplicate employee registration

Employee search

Employee update

Employee deletion

Employee not found

Invalid input

Salary calculation

Database connection failure

Empty mandatory fields

Implement suitable exception handling and evaluate the application with respect to usability, correctness, reliability, database consistency, and maintainability.

Expected Deliverable

Students shall develop a complete working Employee Management and Payroll desktop application using:

Java AWT – Graphical User Interface

Java Event Handling – User interactions

JDBC – Database connectivity

MySQL Workbench / MySQL – Database

OOP Concepts – Classes, objects, encapsulation, inheritance/polymorphism where appropriate

Exception Handling – Application and database errors

SQL – Database operations and queries

The final demonstration should include the running application, database tables, sample records, CRUD operations, payroll calculation, validation, exception handling, and test results.

UPLOAD PDF FILES * (MAX 2 FILES · 10 MB EACH)



File 1

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